

# Why the Digha Tide Program Does Not Exactly Match the Source Data

A simple explanation of the problem, its limitations, and how a future version can solve it

Created by Rishan Bhattacharya

## 1. The Problem We Discovered

We gave the program the genuine Digha Tidechart values from 19 August to 28 August 2026. There are 10 days and 38 reported high- and low-tide points. For example, on 19 August the source says that 02:02 was a high tide of 2.70 m, 08:33 was a low tide of 0.33 m, 14:17 was a high tide of 2.63 m, and 20:55 was a low tide of 0.37 m. When we look at the program's reconstructed curve, however, something strange happens: sometimes its high-tide time is close to the source but its low-tide time is far away; if we make the low tide line up better, the high tide can move away. This is not because the source data are necessarily wrong. It comes mainly from what information our present mathematical fitting method actually uses.

The easiest way to understand the problem is to imagine that a teacher gives a student four points on a graph and says, "These two points are peaks and these two points are valleys." The student draws one smooth mathematical curve through approximately the right heights. But the student's instructions only say, "Be close to these points." They do not say, "Your curve must actually turn around exactly at every peak and valley." The curve can therefore pass near the correct points while its true peaks and valleys occur somewhere else. That is essentially what is happening in our tide program.

## 2. What the Program Is Actually Being Asked to Do

The current program takes each source observation mainly as a time and a height: at a particular time, the water level should be approximately a particular number. It then finds harmonic coefficients by least squares. In simple language, least squares tries to draw the smoothest available combination of the chosen tidal waves that keeps the overall vertical mistakes as small as possible. The program does know that the original data contain a high/low label, because the data are stored with a type value, but that label is not used inside the least-squares fitting calculation.

This is the central limitation. The source tells us more than the fitting equation uses. If the source says "02:02 is HIGH," mathematically that means the water level is not only 2.70 m at 02:02; the curve should also have zero slope there because it is turning from rising to falling. Our current fitting procedure uses the first fact, the height, but not the second fact, the turning-point condition.

**At a high tide we ideally want:  $\eta(t)$  = observed height and  $d\eta/dt = 0$ .**

**At a low tide we similarly want:  $\eta(t)$  = observed height and  $d\eta/dt = 0$ , but the curvature must point upward.**

## 3. A Very Easy Economics Example

Imagine you are estimating a demand curve. You are given two observations: at a price of ₹10, quantity demanded is 100; at a price of ₹20, quantity demanded is 50. Suppose the researcher also tells you that ₹10 is a local maximum in some unusual experiment. If your regression only tries to pass close to the two quantity observations, it may fit both quantities well but it does not automatically

know that the first point must be a turning point. You have used the coordinates of the observation but thrown away part of the economic information attached to it.

The tide problem is the same idea. A tide table does not merely provide numbers such as 2.70 m and 0.33 m. It labels those numbers as HIGH and LOW. That label carries mathematical information about the slope of the underlying curve. Our present program uses the numbers and times but does not put that extra information into the fitting objective.

## 4. Why a Small Height Error Can Still Mean a Large Time Error

This is probably the most confusing part. The program can have a very small height error while still putting the actual maximum or minimum at the wrong time. Think of a smooth hill. If you move the top of the hill a little to the right, the height near the original top may hardly change, especially because the hill is almost flat near its summit. Therefore a curve can pass very close to the correct height at 2:02 without having its mathematical maximum exactly at 2:02.

The same happens at low tide. Near the bottom of a valley the curve is relatively flat. A model may predict almost the same water level at 08:33 but place the actual mathematical minimum at 07:50 or 09:10. So a small vertical error does not guarantee a small horizontal, or timing, error.

That is why our present RMSE is not a tide-time accuracy measure. The fitting error is calculated from the difference between the source height and the model height at the source timestamp. It does not directly calculate how many minutes early or late the model's own high and low points occur.

## 5. Why the Program's Red High/Low Points Can Move

After fitting the harmonic curve, the program performs a different operation. It calculates the derivative of the curve and searches for places where the derivative becomes zero. Those points are declared high or low tides. In other words, the program first says, "Here is the best smooth curve I can fit to the heights," and only afterwards asks, "Where are the mathematical peaks and valleys of this curve?"

Because the fitting stage did not force the source high/low timestamps to be turning points, there is no guarantee that the derivative will become zero at those exact source times. The model's red points are therefore the extrema of the fitted curve, not simply copies of the source tide-table events.

## 6. Limitation One: Only 10 Days of Data

The program is calibrated from only 10 calendar days, from 19 August through 28 August, with 38 extrema. This is enough to see a clear tidal pattern, but it is a very short window for estimating a complicated astronomical process. Tides are made from several waves whose periods are close to one another. Over a short period, some of those waves can look very similar, making it difficult to separate their individual contributions reliably.

A simple analogy is trying to identify two singers from a ten-second recording when both are singing almost the same note. You can reproduce the sound fairly well, but it is difficult to say exactly how much each singer contributed. A longer recording gives the voices more opportunity to separate. Similarly, a longer tide record gives the tidal constituents more opportunity to reveal their different frequencies.

## 7. Limitation Two: We Have Only 38 Turning Points, Not a Continuous Sea-Level Record

A real tide gauge might record the sea level every minute or every five minutes. In a single month that would give thousands of observations. Our dataset contains only the reported high and low points. We therefore know what happened at the turning points, but we do not know the exact shape of the water-level curve between them.

Imagine studying a student's exam performance using only the marks from the first and last question of every section. You can learn something, but you cannot reconstruct every step the student took between those points. The continuous tide curve contains much more information than its turning points alone.

## 8. Limitation Three: Six Constituents Are Still a Simplification

The program selected M2, S2, N2, K1, O1 and M4 from a small set of candidate models. These are physically meaningful constituents, but the real tide at Digha is not made of only six waves. A professional harmonic analysis can involve many more constituents. However, adding many constituents to only 38 observations would create another problem: too many parameters relative to the amount of information available.

So the program faces a trade-off. Too few waves and the curve may be too simple. Too many waves and the model may start learning accidental details of the short dataset rather than a stable tidal pattern. This is the same bias-versus-variance problem economists and statisticians face when choosing between an overly simple model and an overly complicated one.

## 9. Limitation Four: Height Fit and Timing Fit Are Different Objectives

The current program's validation mainly asks, "How close is the predicted height to the source height at the known timestamps?" But for a tide table we also care about, "How many minutes away is the predicted high tide?" and "How many minutes away is the predicted low tide?" These are different measures of performance.

For example, a prediction of 2.70 m at 02:02 when the source says 2.70 m at 02:02 could have zero height error. But if the program's actual maximum occurs at 02:35, it has a 33-minute timing error. The present RMSE would not show that 33-minute error because the RMSE is not based on the difference between the two event times.

## 10. Limitation Five: The Source Itself Is a Tide Table

There is another important layer. The 19–28 August values came from the Tidechart table shown in the supplied source. They are valid source data for our project, but they are not the same thing as a raw, continuous Digha tide-gauge observation record. Therefore the program is essentially learning the pattern represented by that published table. It is not independently measuring the ocean and discovering the true local harmonic constants from scratch.

This is why the program should be called an empirical harmonic extrapolation. Its purpose at this stage is educational and experimental: to take a short set of real-world tidal extrema and explore how a harmonic model can reproduce and extend their pattern.

## 11. Limitation Six: Extrapolation Is Harder Than Reconstruction

The program can fit the August data quite closely because those observations were used to estimate its coefficients. But October 12 is far outside the calibration window. The farther we move from the

observations, the more important small errors in phase and constituent amplitudes become. A model can look excellent inside its training period and still drift in the future.

This is similar to an economic forecasting model. Suppose GDP data for ten quarters are fitted very well. That does not automatically mean the model will forecast GDP accurately five years later. The model may have captured the recent pattern but missed a structural feature that becomes important later. Tide prediction has an analogous problem when the calibration record is very short.

## 12. So What Exactly Is Going Wrong?

Putting everything together, the mismatch has a clear chain of causes. First, we have only ten days of sparse extrema. Second, the harmonic model has to estimate several wave components from that small sample. Third, least squares currently tries to match the heights at the known times but does not enforce the fact that a HIGH must be a mathematical maximum and a LOW must be a mathematical minimum. Fourth, the program finds its own maxima and minima afterwards by setting the derivative to zero. Finally, the model is extrapolated well beyond the short calibration window.

So the issue is not simply "the equation is wrong." The equation is a reasonable harmonic equation. The problem is that the available data and the fitting objective do not contain enough information, or do not use all the information available, to force the model's turning points to coincide with the source turning points.

## 13. How We Can Solve It in the Future

The first improvement should be to acquire a continuous Digha tide-gauge record. Instead of 38 points over ten days, we should ideally obtain at least 30 days of continuous observations as a minimum experimental dataset, and preferably several months or a full year. A longer record would allow the program to distinguish tidal constituents much more reliably and would provide enough information to test whether the fitted constants remain stable.

The second improvement is to use the high/low information properly if we continue working with extrema. For every source high tide, the future fitting method should try to satisfy both the observed height and the condition that the derivative is zero there. For every source low tide, it should do the same while requiring the correct curvature. This would make the fitting problem explicitly aware of tide-event timing rather than treating every source point as an ordinary height observation.

The third improvement is to measure timing accuracy separately. After fitting the model, we should compare every predicted high tide with the corresponding observed high tide and every predicted low tide with the corresponding observed low tide. The results should include height RMSE, mean absolute height error, high-tide timing error in minutes, low-tide timing error in minutes, and maximum timing error. This would tell us whether the program is actually getting the tide times right.

The fourth improvement is to use a longer and more complete harmonic model only when the data justify it. With months or a year of continuous observations, we could gradually add additional constituents, calculate proper astronomical arguments and nodal corrections, and test whether each additional constituent improves independent forecasts rather than merely improving the fit to the training data.

Finally, the improved system should be tested on data it has never seen. For example, we could fit the model using January through September and then predict October. Only after comparing the October predictions with independent Digha observations would we be able to make a meaningful statement about real-world predictive accuracy.

## 14. The Future Version in One Simple Story

Think of our present program as a student who has been given ten days of a very difficult exam subject and is asked to reconstruct the teacher's entire question paper. The student has enough information to notice the main pattern, but not enough to know every detail. The student also knows the marks obtained at important points, but has not been told to reproduce every turning point exactly. That is why the answer looks broadly right while some details are shifted.

Now imagine giving the student a full year's worth of notes, every intermediate step, and the teacher's exact rules. The student can separate the different patterns, check the answer against unseen questions, and discover where the model fails. That is what we want to do with Digha: more continuous data, a better harmonic basis, explicit high/low timing constraints, proper astronomical calculations, and independent validation.

## 15. Final Conclusion

The mismatch between the 19–28 August source tide times and the program's displayed high/low times is therefore understandable. The current program is not simply copying the tide table; it is fitting a smooth six-constituent harmonic curve to the reported heights and then finding the extrema of that curve. Because the fitting step does not explicitly force the source high and low timestamps to be turning points, the model's own extrema can occur earlier or later than the source events.

The short ten-day, 38-extrema dataset makes the problem more difficult because the program has limited information with which to separate several tidal frequencies. The solution is not to abandon the harmonic approach. The better solution is to give the harmonic approach better data and a better fitting objective: continuous observations, longer calibration periods, explicit event-timing constraints when using extrema, more complete astronomical calculations, separate timing and height validation, and independent future-period testing.

In one sentence: our current program learns the heights at known tide times; the future program should learn the entire tide curve and the exact turning points that generate those high and low tides.